The Quantum Hall Effect

Self-Project Report

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Abstract

This report provides a comprehensive and extensive study of the Quantum Hall Effect, synthesizing the advanced concepts covered in David Tong's lecture notes. It explores both the Integer and Fractional Quantum Hall Effects in high detail, delving into the underlying quantum mechanics of particles in magnetic fields, the emergence of Landau levels, and the critical roles of disorder, topology, and gauge invariance. Furthermore, it extensively examines the interacting electron picture, describing the Laughlin states, fractional charge, anyonic statistics, the hierarchy of states, composite fermions, and the effective field theories governed by Chern-Simons dynamics. The discussion is broadened to include non-Abelian states and edge mode formalisms, providing a complete physical and mathematical overview of this topological phase of matter.

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Chapter 1

Introduction and The Classical Hall Effect

1.1 Introduction

The quantum Hall effect represents one of the most profound discoveries in condensed matter physics, fundamentally altering our understanding of phases of matter. By restricting electrons to a two-dimensional plane and subjecting them to a strong perpendicular magnetic field at low temperatures, the system exhibits macroscopic quantum phenomena that defy classical expectations. The Hall conductivity becomes exactly quantised in integer or specific rational multiples of the fundamental constant e^2/h.

This quantisation is not merely a consequence of the microscopic details of the material but is deeply rooted in topology. The integer quantum Hall effect (IQHE) highlights the role of topological invariants, specifically Chern numbers, in non-interacting electron systems. In contrast, the fractional quantum Hall effect (FQHE) is driven by strong Coulomb interactions between electrons, leading to the emergence of highly correlated liquid states characterized by fractional charges and anyonic statistics.

1.2 The Classical Hall Effect

The classical Hall effect, discovered by Edwin Hall in 1879, is a direct consequence of the Lorentz force acting on moving charges. Consider a two-dimensional electron gas (2DEG) confined to the $x-y$ plane in the presence of a uniform magnetic field $\mathbf{B} = B\hat{z}$.

An electric field $\mathbf{E} = E\hat{x}$ drives a current in the x-direction. The Lorentz force deflects the electrons, leading to an accumulation of charge on the boundaries and establishing a transverse electric field E_y, known as the Hall field.

The equation of motion for an electron of mass m and charge $-e$ in the Drude model is given by:

$$m\frac{d\mathbf{v}}{dt} = -e\mathbf{E} - e\mathbf{v} \times \mathbf{B} - \frac{m\mathbf{v}}{\tau} \tag{1.1}$$

where τ is the momentum relaxation (scattering) time. In the steady state ($d\mathbf{v}/dt = 0$), the current density $\mathbf{J} = -ne\mathbf{v}$ obeys Ohm's law $\mathbf{J} = \sigma\mathbf{E}$. The conductivity tensor σ is expressed as:

$$\sigma = \frac{\sigma_{DC}}{1 + \omega_B^2\tau^2} \begin{pmatrix} 1 & -\omega_B\tau \\ \omega_B\tau & 1 \end{pmatrix} \tag{1.2}$$

Here, $\omega_B = eB/m$ is the cyclotron frequency, and $\sigma_{DC} = ne^2\tau/m$ is the standard Drude DC conductivity. The resistivity tensor, defined as the inverse of the conductivity tensor ($\rho = \sigma^{-1}$), is:

$$\rho = \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ -\rho_{xy} & \rho_{yy} \end{pmatrix} = \frac{1}{\sigma_{DC}} \begin{pmatrix} 1 & \omega_B\tau \\ -\omega_B\tau & 1 \end{pmatrix} = \begin{pmatrix} \frac{m}{ne^2\tau} & \frac{B}{ne} \\ -\frac{B}{ne} & \frac{m}{ne^2\tau} \end{pmatrix} \tag{1.3}$$

A key observation from classical physics is that the transverse (Hall) resistivity $\rho_{xy} = B/ne$ is strictly proportional to the magnetic field B and is completely independent of the scattering time τ. It depends solely on the charge carrier density n and fundamental constants. The classical prediction states that plotting ρ_{xy} against B should yield a straight line.

Chapter 2

Quantum Mechanics of Particles in a Magnetic Field

2.1 Landau Levels

The breakdown of the classical linear expectation occurs at high magnetic fields and low temperatures, necessitating a quantum mechanical treatment. The Hamiltonian for a single spinless electron in a magnetic field is:

$$H = \frac{1}{2m}(\mathbf{p} + e\mathbf{A})^2 \tag{2.1}$$

where $\mathbf{p} = -i\hbar\nabla$ is the canonical momentum, and $\mathbf{A}$ is the vector potential such that $\nabla\times\mathbf{A} = \mathbf{B}$. The mechanical momentum $\boldsymbol{\pi} = \mathbf{p}+e\mathbf{A}$ is gauge invariant but its components do not commute:

$$[\pi_x, \pi_y] = -i\hbar eB \tag{2.2}$$

This commutation relation is reminiscent of the position and momentum operators, allowing us to define raising and lowering operators:

$$a = \frac{1}{\sqrt{2e\hbar B}}(\pi_x - i\pi_y), \quad a^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\pi_x + i\pi_y) \tag{2.3}$$

which satisfy $[a, a^\dagger] = 1$. The Hamiltonian then takes the form of a one-dimensional quantum harmonic oscillator:

$$H = \hbar\omega_B \left(a^\dagger a + \frac{1}{2} \right) \tag{2.4}$$

The energy spectrum consists of equally spaced discrete levels, known as Landau levels:

$$E_n = \hbar\omega_B \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \ldots \tag{2.5}$$

2.2 Gauge Choices and Wavefunctions

The specific form of the wavefunctions depends on the choice of gauge.

2.2.1 Landau Gauge

Choosing the Landau gauge $\mathbf{A} = xB\hat{y}$ preserves translational invariance in the y-direction. We can use the separation of variables ansatz $\psi_k(x, y) = e^{iky} f_k(x)$. The resulting 1D Schrödinger equation for $f_k(x)$ is a harmonic oscillator shifted by $x_0 = -kl_B^2$, where $l_B = \sqrt{\hbar/eB}$ is the magnetic length. The wavefunctions are parallel strips localized in x and extended in y.

2.2.2 Symmetric Gauge

The symmetric gauge $\mathbf{A} = \frac{B}{2}(-y, x)$ preserves rotational symmetry around the origin, making angular momentum a good quantum number. Introducing complex coordinates $z = x - iy$ and $\bar{z} = x + iy$, the lowest Landau level (LLL) wavefunctions ($n = 0$) are analytic functions of z multiplied by a Gaussian envelope:

$$\psi_{LLL,m}(z, \bar{z}) \propto z^m e^{-|z|^2/4l_B^2} \tag{2.6}$$

where $m \geq 0$ is the angular momentum quantum number. These wavefunctions are localized on rings of radius $r_m = \sqrt{2m} l_B$.

2.3 Degeneracy of Landau Levels

Each Landau level exhibits a massive macroscopic degeneracy. In a sample of area A, the number of available states per Landau level is:

$$\mathcal{N} = \frac{eBA}{2\pi\hbar} = \frac{\Phi}{\Phi_0} \tag{2.7}$$

where $\Phi = BA$ is the total magnetic flux through the sample, and $\Phi_0 = h/e$ is the magnetic flux quantum.

Chapter 3

The Integer Quantum Hall Effect (IQHE)

3.1 Observation and Idealized Conductivity

Discovered by von Klitzing in 1980, the IQHE demonstrates that ρ_{xy} develops extended plateaux at exactly:

$$\rho_{xy} = \frac{h}{e^2\nu} \tag{3.1}$$

where ν is an integer. Concurrently, the longitudinal resistivity ρ_{xx} drops to zero. This occurs when the Fermi energy lies in the gap between the ν-th and $(\nu+1)$-th Landau levels. If we define the filling fraction as the ratio of electron density to the density of states per Landau level, $\nu = n\Phi_0/B$, an integer filling trivially yields the correct macroscopic conductivity. However, this alone does not explain the existence of the plateaux over a range of magnetic fields.

3.2 The Crucial Role of Disorder

In a perfect crystal, varying B slightly would result in a partially filled Landau level, destroying the zero longitudinal resistance. The plateaux are physically stabilized by disorder (impurities). The random potential of impurities broadens the perfectly sharp Landau levels into energy bands.

Crucially, the states in the tails of these bands are Anderson localized—they are trapped in potential valleys or peaks and cannot traverse the sample. Only a narrow window of states near the center of the band remains extended (delocalized). As the magnetic field is varied, the Fermi energy sweeps through the localized states. Because adding or removing electrons from localized states does not contribute to macroscopic charge transport, the conductivity tensors remain rigorously constant, thus forming the plateaux. The extended states, astonishingly, carry extra current to perfectly compensate for the localized ones.

3.3 Laughlin's Gauge Invariance Argument

Laughlin provided an elegant argument showing that the quantization must be exact. Consider the 2DEG mapped onto a Corbino ring (an annulus) with a magnetic flux Φ threaded through the central hole. This flux does not touch the 2DEG.

Varying this flux induces an azimuthal electric field by Faraday's law. Because the system has a gap (the Fermi level is in the mobility gap of localized states), the Hamiltonian must return to itself up to a gauge transformation when Φ changes by one flux quantum $\Phi_0 = h/e$. This process pumps exactly an integer number of electrons, ν, from the inner edge to the outer edge of the annulus. The transfer of ν electrons for every flux quantum threaded immediately yields the exact quantization of the Hall conductance: $\sigma_{xy} = \nu e^2/h$.

3.4 Topology and the TKNN Invariant

The ultimate explanation for the exactness of the IQHE lies in topology. Thouless, Kohmoto, Nightingale, and den Nijs (TKNN) applied the Kubo formula for conductivity to a lattice model in a magnetic field. They proved that the Hall conductivity is intrinsically linked to the Berry curvature of the Bloch bands.

The Kubo formula for the transverse conductivity can be rewritten as:

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int_{\mathrm{BZ}} \frac{d^2k}{(2\pi)^2} \mathcal{F}_{xy}(\mathbf{k}) = \frac{e^2}{h} \mathcal{C} \tag{3.2}$$

where $\mathcal{F}_{xy}(\mathbf{k})$ is the Berry curvature in momentum space, and the integral over the entire magnetic Brillouin Zone (BZ) is a topological invariant known as the first Chern number, $\mathcal{C}$. Because $\mathcal{C}$ must be an integer, σ_{xy} is robustly quantised. It cannot change under continuous deformations of the system as long as the mobility gap remains open.

Chapter 4

The Fractional Quantum Hall Effect (FQHE)

In 1982, Tsui, Störmer, and Gossard discovered plateaux at fractional filling fractions, most notably at $\nu = 1/3$. Because the lowest Landau level is only partially filled, there is a massive ground state degeneracy in the non-interacting picture. This degeneracy is lifted by the repulsive Coulomb interactions between electrons, which drive the formation of a novel, strongly correlated many-body state.

4.1 The Laughlin Wavefunction

Solving the many-body interacting Hamiltonian analytically is practically impossible. Instead, Robert Laughlin guessed a trial wavefunction for filling fractions $\nu = 1/m$ (where m is an odd integer for fermions). The wavefunction is built from the LLL symmetric gauge single-particle states:

$$\psi_m(z_1, \ldots, z_N) = \prod_{i<j}^{N} (z_i - z_j)^m e^{-\sum_{i=1}^{N} |z_i|^2/4l_B^2} \tag{4.1}$$

This wavefunction elegantly builds in the Pauli exclusion principle. By setting $m = 3$, the wavefunction forces electrons to avoid each other even more strongly than required for simple fermions (where $m = 1$), significantly minimizing the Coulomb repulsion energy.

4.2 The Plasma Analogy and Incompressibility

To compute expectation values, Laughlin introduced the plasma analogy. The probability density $|\psi_m|^2 = e^{-\beta U}$ is mathematically mapped to the Boltzmann distribution of a 2D classical one-component plasma. The effective potential is:

$$U(z_1, \ldots, z_N) = -m^2 \sum_{i<j} \ln |z_i - z_j| + \frac{m}{4l_B^2} \sum_{i=1}^{N} |z_i|^2 \tag{4.2}$$

with an artificial inverse temperature $\beta = 2/m$. This maps the problem to plasma particles of charge $q = m$ immersed in a uniform neutralizing background.

The requirement of macroscopic charge neutrality in the plasma guarantees that the electron density is rigorously locked to $n = 1/(2\pi l_B^2 m)$, precisely the density for a $\nu = 1/m$ state. The plasma perfectly screens density fluctuations, implying that the quantum Hall fluid is an incompressible liquid. A finite energy gap exists for creating density waves or localized excitations.

4.3 Fractional Charge of Quasi-Holes

Excitations in the Laughlin state are created by adiabatic insertion of flux quanta. A quasi-hole at position η is described by the wavefunction:

$$\psi_{\text{hole}}(z_i, \eta) = \prod_{i=1}^{N} (z_i - \eta) \psi_m(z_i) \tag{4.3}$$

This operator creates a zero of the wavefunction at η for all electrons. In the plasma analogy, this acts as a phantom charge of 1, repelling the plasma particles. Because the plasma particles carry charge m, the deficit of real electrons corresponds to an effective charge of exactly $1/m$. Therefore, the elementary excitations of the FQHE fluid carry a fractional charge $e^* = e/m$.

4.4 Fractional Statistics (Anyons)

In three spatial dimensions, particles are strictly bosons or fermions because swapping them twice is topologically equivalent to not swapping them at all. However, in two dimensions, swapping particles twice is topologically equivalent to one particle winding completely around another (braiding). The path cannot be continuously shrunk to a point without the particles colliding.

Because of this 2D topology, identical particles can acquire any phase $e^{i\theta}$ upon exchange. Such particles are termed "anyons." For the quasi-holes in the Laughlin state, computing the Berry phase of dragging one quasi-hole around another yields a statistical parameter:

$$\alpha = \frac{1}{m} \tag{4.4}$$

This indicates that the fundamental excitations of the FQHE are anyons, obeying fractional statistics intermediate between bosons and fermions.

Chapter 5

Hierarchy States and Composite Fermions

5.1 The Hierarchy Construction

The Laughlin wavefunction explains states at $\nu = 1/m$. To explain other observed fractions (like $2/5, 3/7, \dots$), Haldane and Halperin proposed a hierarchy scheme. Just as electrons condense into a Laughlin liquid, the quasi-particles themselves can condense into their own Laughlin-like liquid if their density is sufficiently high. This process can be iterated, leading to a continued fraction expression for the allowed filling fractions, which successfully predicts the observed odd-denominator plateaux.

5.2 Jain's Composite Fermions

An alternative, immensely powerful physical picture was proposed by Jainendra Jain. He introduced the concept of the "composite fermion," which is an electron bound to an even number of magnetic flux quanta (vortices).

By attaching $2p$ vortices to each electron (usually $2p = m - 1$), the electrons effectively "absorb" part of the external magnetic field. The composite fermions then behave as weakly interacting particles experiencing a much weaker effective magnetic field:

$$B^* = B - 2p\Phi_0 n \tag{5.1}$$

When these composite fermions completely fill an integer number ν^* of their effective Landau levels (referred to as Λ-levels), the original electrons exhibit a FQHE. The relation between the electron filling fraction ν and the composite fermion integer filling ν^* is:

$$\nu = \frac{\nu^*}{2p\nu^* \pm 1} \tag{5.2}$$

This magnificent mapping completely maps the fractional quantum Hall effect onto the integer quantum Hall effect of composite fermions. Setting $2p = 2$ and $\nu^* = 1, 2, 3 \ldots$ gives the principal sequence $\nu = 1/3, 2/5, 3/7 \ldots$, precisely mirroring experimental data.

5.2.1 The Half-Filled Landau Level

A spectacular prediction of the composite fermion theory occurs at $\nu = 1/2$. Here, $B^* = 0$. The composite fermions experience zero net magnetic field and thus form a compressible Fermi liquid rather than a gapped quantum Hall state. This brilliantly explains the distinct absence of a Hall plateau at $\nu = 1/2$.

Chapter 6

Non-Abelian Quantum Hall States

While the vast majority of FQHE states occur at odd denominators, an extremely prominent state is observed at the even denominator fraction $\nu = 5/2$.

6.1 The Moore-Read Pfaffian State

To explain the $\nu = 5/2$ state (which is effectively $\nu = 1/2$ in the second Landau level), Moore and Read proposed a wavefunction based on a BCS-like pairing of composite fermions. The spatial part of the pairing is described by a Pfaffian:

$$\psi_{\mathrm{MR}} = \mathrm{Pf}\left(\frac{1}{z_i - z_j}\right) \prod_{i<j} (z_i - z_j)^2 e^{-\sum |z_i|^2/4l_B^2} \tag{6.1}$$

This state represents a topological $p + ip$ superconductor of composite fermions.

6.2 Majorana Zero Modes and Topological Quantum Computing

The excitations of the Moore-Read state are profoundly different from the Laughlin states. The quasi-holes carry a charge of $e/4$. Furthermore, each quasi-hole traps a Majorana zero mode in its core.

When multiple quasi-holes are present, these Majorana modes give rise to a ground state degeneracy that grows exponentially as 2^{n-1} for $2n$ quasi-holes. Braiding these quasi-holes around each other performs non-trivial unitary rotations within this degenerate ground state subspace. Because these matrices do not commute, the statistics are Non-Abelian.

The non-local storage of quantum information in the degenerate subspace and the topological nature of the braiding operations make the $\nu = 5/2$ state the leading candidate for realizing fault-tolerant Topological Quantum Computing.

Chapter 7

Effective Field Theories: Chern-Simons Dynamics

To capture the macroscopic, low-energy universal features of the quantum Hall effect, we construct Topological Field Theories, specifically Chern-Simons theory in 2+1 dimensions.

7.1 Abelian Chern-Simons Theory

For the Laughlin $\nu = 1/m$ state, we introduce an emergent dynamical $U(1)$ gauge field a_μ. The effective action is composed of a Chern-Simons term for a_μ and a coupling to the external electromagnetic field A_μ:

$$S = \int d^3x \left(\frac{1}{2\pi} \epsilon^{\mu\nu\rho} A_\mu \partial_\nu a_\rho - \frac{m}{4\pi} \epsilon^{\mu\nu\rho} a_\mu \partial_\nu a_\rho \right) \tag{7.1}$$

Integrating out the dynamical field a_μ yields an effective action for the external field:

$$S_{\text{eff}}[A] = \frac{1}{4\pi m} \int d^3x \epsilon^{\mu\nu\rho} A_\mu \partial_\nu A_\rho \tag{7.2}$$

The current is simply $J^\mu = \delta S / \delta A_\mu = \frac{1}{2\pi m} \epsilon^{\mu\nu\rho} \partial_\nu A_\rho$. The spatial components of this equation directly yield the Hall conductivity $\sigma_{xy} = \frac{e^2}{h} \frac{1}{m}$. The Chern-Simons framework effortlessly encapsulates the quantized Hall transport, the fractional charge of quasi-particles (via equations of motion with source currents), and their anyonic fractional statistics.

Chapter 8

Edge Modes and the Bulk-Boundary Correspondence

While the bulk of a quantum Hall fluid is gapped and insulating, the edges harbor gapless, conducting modes. The classical skipping orbits of electrons at the boundaries translate quantum mechanically into chiral, one-way propagating modes.

8.1 The Chiral Luttinger Liquid

For the Laughlin state, the boundary dynamics are described by a 1+1 dimensional Chiral Boson CFT, formulated by Xiao-Gang Wen. The action for the bosonic field $\phi(x,t)$, representing edge density fluctuations, is:

$$S_{\text{edge}} = \frac{m}{4\pi} \int dtdx \left(\partial_t \phi \partial_x \phi - v(\partial_x \phi)^2 \right) \tag{8.1}$$

This describes a chiral Luttinger liquid. The electron operator on the edge is identified as $\Psi_e^\dagger \propto e^{im\phi}$, which has a scaling dimension $m/2$. This non-trivial scaling dimension leads to a striking prediction for tunneling experiments: the differential conductance $G \sim V^{2m-2}$ violates Ohm's law, exhibiting a profound non-linear scaling that has been beautifully confirmed by experiment.

8.2 Bulk-Boundary Correspondence

The gapless edge modes and the topological bulk are inextricably linked. The boundary Conformal Field Theory (CFT) is the holographic dual of the bulk Chern-Simons theory. Moore and Read famously demonstrated that the bulk ground state wavefunctions can be exactly constructed as the spatial correlation functions of vertex operators in the boundary CFT. This profound bulk-boundary correspondence continues to inspire advances across condensed matter, high energy theory, and string theory.

Chapter 9

Conclusion

The Quantum Hall Effect is a cornerstone of modern condensed matter physics. It has shattered the traditional Landau paradigm of classifying phases of matter by local symmetry breaking. Instead, it has introduced the world to topological order—a paradigm where phases are defined by ground state degeneracy, topological invariants, and fractionalized excitations.

From the exactness of the TKNN Chern numbers to the beautiful synthesis of composite fermions and the futuristic promise of non-Abelian anyons for quantum computation, the study of the quantum Hall effect bridges experimental reality with some of the most elegant mathematical concepts in theoretical physics, including Chern-Simons theory, conformal field theory, and topology.